ecules called polymers, much like hundreds of paper clips strung together.

The fact that it can be moulded into any shape when heated adds to the material's versatility. Based on their behaviour when heated, plastics can be divided into two broad categories—thermoplastics, which become soft upon heating, can be melted and reshaped 2-3 times, and are therefore recyclable; and thermosets, which can only be moulded once after heating, and thus cannot be recycled. The first include the familiar PET (polyethylene terephthalate) water bottles, PVC pipes, polypropylenes that are used in disposable cups and food containers, and the polyethylene used in shopping bags etc. (see *Waste to Wealth*). Thermosets include bakelite, melamine and vulcanised rubber found in shoe soles, car tyres, rubber hoses etc.

Lightweight, durable, waterproof and inexpensive to boot, it was a matter of time before plastic began to be used across industries—from food and fashion to construction, automotive and healthcare—as well as in homes, an indispensable accessory when you went out shopping, whether for groceries or for garments. As the population and economy boomed, so did plastic manufacture and use. The plastic industry, estimated to be Rs 3.5 lakh crore in FY23, is projected to grow to Rs 10 lakh crore in FY28. It is home to 30,000 processing units, a majority of them micro, small and medium enterprises, employing more than 4 million people and contributing 1.4 per cent to the GDP. In FY25, India exported plastics worth $12.5 billion (approx. Rs 1.07 lakh crore), according to the Plastics Export Promotion Council.

Today, India is among the top three consumers of plastics globally, after China and the US, though its per capita plastic consumption of 13 kg is less than half the global average of 30 kg. But while China generates 0.08 kg per person per day of plastic waste, one and a half times more than the 0.054 kg generated in India, our inability to manage the waste makes us the largest plastic emitters in the world, more than three times China's emissions of 2.8 Mt, according to the calculations in the *Nature* study.

PLASTIC AS HAZARD

Its very ubiquity has made plastic a global menace, in which India is said to play a major part. Plastic's non-biodegradable nature means it stays around forever, and everywhere.

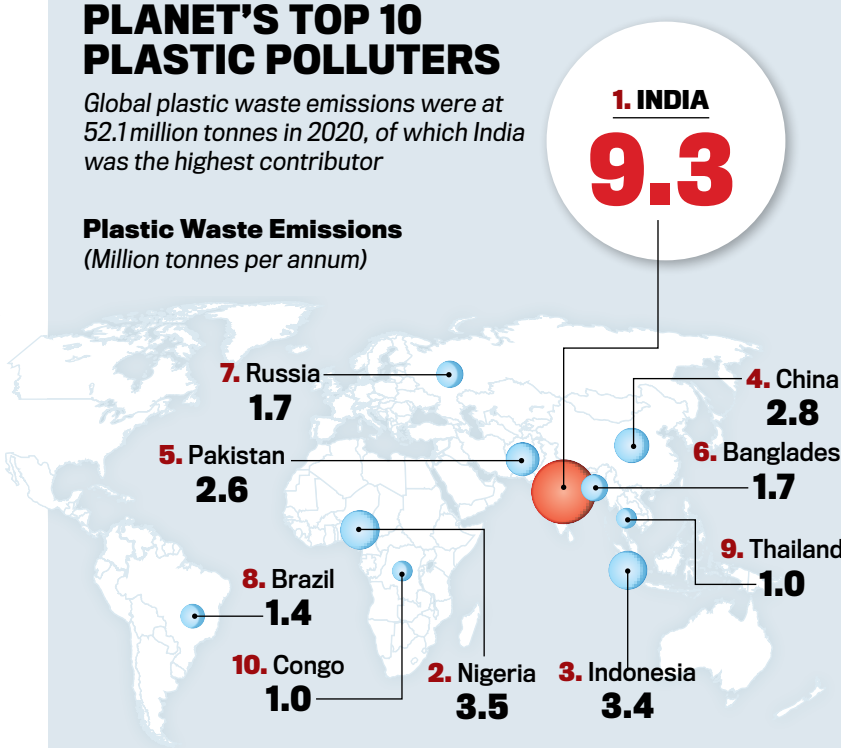
DUBIOUS DISTINCTION

India may be producing less plastic waste per capita than, say, China, but mismanagement of that waste makes it the world's worst offender in plastic emissions

PLANET'S TOP 10 PLASTIC POLLUTERS

Global plastic waste emissions were at 52.1 million tonnes in 2020, of which India was the highest contributor

Plastic Waste Emissions
(Million tonnes per annum)

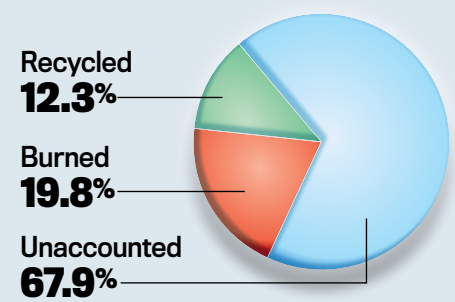


Source: Nature, Sept. 2024 (data for 2020)/Angeliki Savvantoglou of Bear Bones / phys.org

LITTERING IS THE MAIN SOURCE FOR EMISSIONS IN THE GLOBAL NORTH WHEREAS UNCOLLECTED WASTE IS THE LARGEST SOURCE ACROSS THE GLOBAL SOUTH

THE PLASTIC WASTE CRISIS

The lion's share of plastic waste disposal in India is unaccounted-for, lingering in the environment



Source: A 2022 analysis by Centre for Science and Environment